

AADK 3 - Task 2 — Image Inclusion & Resource Management in Compose

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1 What is the Image Composable?

The **Image** composable in Jetpack Compose is used to display images in a UI. It can load images from:

- Drawable resources (painterResource)
- URLs (using libraries like Coil)
- Bitmap or vector assets

Example Use-Case:

Displaying a banner image at the top of a profile screen or showing a product image in an e-commerce app.

Example structure:

- Column
 - Image
 - Text
 - Button

The Image composable allows control over:

- contentScale (Crop, Fit, FillBounds)
- modifier (size, padding, fillMaxWidth)
- alignment

2 Why Image Density Matters in Android

Android devices have different screen densities (mdpi, hdpi, xhdpi, xxhdpi, etc.). If images are not optimized for density:

- Images may appear **blurry** on high-density screens.
- Images may look **too large or too small**.
- App size may unnecessarily increase.
- Performance can be affected.

drawable-nodpi

The drawable-nodpi folder stores images that should NOT be scaled automatically based on screen density (e.g., large background images, illustrations).

If density management is ignored:

- UI becomes inconsistent across devices.
- Users may experience distorted visuals.
- Layout alignment may break.

3 Best Practices for Adding Images in Compose

✓ 1. Use Proper Naming Conventions

- Use lowercase letters.
- Use underscores (e.g., profile_banner.png).
- Avoid spaces and special characters.

This improves maintainability and prevents build errors.

✓ 2. Optimize Image Size

- Compress images before importing.
- Avoid using very large images for small UI elements.
- Prefer vector drawables (.xml) for icons.

This reduces APK size and improves performance.

Visual Storyboard

Frame 1 — Placeholder Text Only

Welcome Title	
This is some paragraph text explaining content	

Composable Structure:

- Column
 - Text (Title)
 - Text (Paragraph)

Layout is simple and vertically arranged.

Frame 2 — Image Added at the Top

[IMAGE]	
Welcome Title	
This is some paragraph	

Composable Structure:

- Column

- Image
- Text (Title)
- Text (Paragraph)

The image pushes text downward and changes visual hierarchy.

Frame 3 — Image Full Width (Cropped)

	[FULL WIDTH CROPPED IMAGE]	
	Welcome Title	
	This is some paragraph	

Composable Structure:

- Column
 - Image(modifier = fillMaxWidth(), contentScale = Crop)
 - Text (Title)
 - Text (Paragraph)

Here:

- fillMaxWidth() makes image span full width.
- contentScale = Crop ensures image fills space without distortion.

This changes visual impact and makes the screen look more modern and structured.

5 Reflection

Images improve visual appeal but introduce challenges. Large images can slow down performance and increase memory usage. Improper scaling may break layouts on different devices. Accessibility must also be considered by providing content descriptions for screen readers. Therefore, images must be optimized, properly placed, and carefully managed in Compose UI design.